

WRITING

Find and describe a painting in the Harvard Art Museum where color is used more than decoratively, where the artist makes careful and deliberate use of color to compose the image.

WATCHING

[Watch](#) a TedX talk about the digital restoration of the Harvard Murals.

READING

- Bevil R. Conway, Saima Malik-Moraleda, and Edward Gibson. Color appearance and the end of hering's opponent-colors theory. *Trends in cognitive sciences*, 27(9):791–804, 2023. ISSN 1364-6613 [Download paper](#)

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- E. H. Gombrich. *The Story of Art*. Phaidon Press, Ltd., London, 16th ed., expanded and redesigned edition, 1995
- Josef Albers. *Interaction of color*. Yale University Press, New Haven [Conn.], new complete ed. edition, 2009. ISBN 9780300146936
- Margaret Livingstone. *Vision and Art: The Biology of Seeing*. Harry N. Abrams, Inc., Publishers, New York, 2002. ISBN 0-8109-0406-3



Figure 191: *Cyan Kitchenette* by Susan Lichtman, 2024

WEEK NINE - SHADOW AND SILHOUETTE

Thursday, 3 April 2025, 12:45 - 2:15 PM EST.

Harvard Art Museums 0600

SILHOUETTES PAINTED IN THE SHADOWS were beginnings of Western Art. The earliest European Art are Paleolithic paintings in the Caves of Altamira in Northern Spain. Made 36,000 years ago, but discovered in 1868, the paintings are charcoal and polychrome profiles of bison and other animals. Profiles – silhouettes painted in the shadows – are just enough to convey meaning through representation. Their original purpose is unknown, perhaps they were celebrations or prayers about hunting.



Figure 192: Great Hall of Polychromes of Altamira, published in 1880



Figure 193: Bison Polychrome, Altamira

Updated: April 7, 2025

GREEK HISTORY gave shadow a role in the beginnings of representational art. Pliny the Elder (AD 23/24 - 79) recorded a story about the origin of portrait sculpture in his *Natural History*. Butades, a potter in Ancient Corinth, noticed a shadow made by lamplight, which his daughter had traced of her boyfriend. Butades filled the outline with clay, inventing the first portrait sculpture.

SHADOW TRACING is an easy-to-use optical effect to help make a realistic representation. This story of Butades was a common theme in European painting, an origin story about Art that reveals a continuing fascination for Greek and Roman antiquity in the 18th and 19th centuries.

PROFILE PORTRAITS of famous Greeks and Romans were most widely seen and most widely distributed on coins. The side profiles is not far from the shadow silhouettes. Side profiles – gaining currency from antiquity and nobility – became an ennobling form of representation.

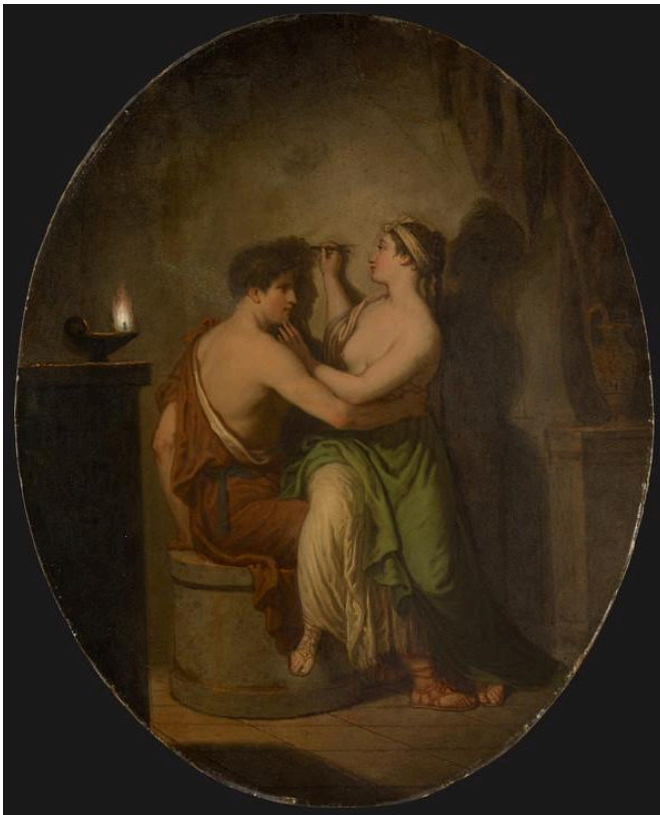


Figure 194: Alexander the Great, 336-323 BCE. [Harvard Art Museums](#)



Figure 195: Caesar Augustus, Silver coin, 19-18BC. [British Museum](#)



Figure 196: Head of Brutus, Roman Republic, 43 -42 BC. Commemorative coin for Julius Caesar's assassination. [British Museum](#)

Figure 197: David Allan, *The Origin of Painting*, 1775. [Link to painting.](#)

ITALIAN RENAISSANCE artists used shadow, like they used perspective, to add depth. Both modeling shadows and cast shadows helped with the illusion of three-dimensionality. Leonardo da Vinci thought a lot about shadows, devoting 67 pages of his *Codex Urbinas* to their study. But in his notes on painting, *Trattato della Pittura*, he advocated leaving them out.

Light too conspicuously cut off by shadows is exceedingly disapproved of by painters. Hence, to avoid such awkwardness when you depict bodies in open country, do not make your figures appear illuminated by the sun, but contrive a certain amount of mist or of transparent cloud to be placed between the object and the sun and thus, – since the object is not harshly illuminated by the sun – the outlines of the shadows will not clash with the outlines of the lights.

A *SALVATOR MUNDI*, painted in 1500 and attributed to Leonardo, uses shadows for modeling. Its light is diffuse and ambiguous. If this painting were by Leonardo, he followed his own advice. Its attribution to Leonardo made it the most expensive painting ever sold at auction, \$450.3 million to Prince Badr bin Abdulla Al Saud in 2017. It has not been seen since.

ANDREA PREVITALI painted a *Salvator Mundi* in 1519. A commitment to realism is in the strong shadow cast by his raised right hand. Another realistic detail is specular reflection from the glass orb. Leonardo placed a similar orb in Christ's left hand. But both Previtali and Leonardo decline optical realism in the orb itself. Instead of painting an optically-distorting but optically-accurate glass lens, both paint transparent windowpanes.



Figure 198: *Salvator Mundi*, 1500, attributed to Leonardo da Vinci and workshop.



Figure 199: *Salvator Mundi*, 1519, Andrea Previtali. [National Gallery](#)

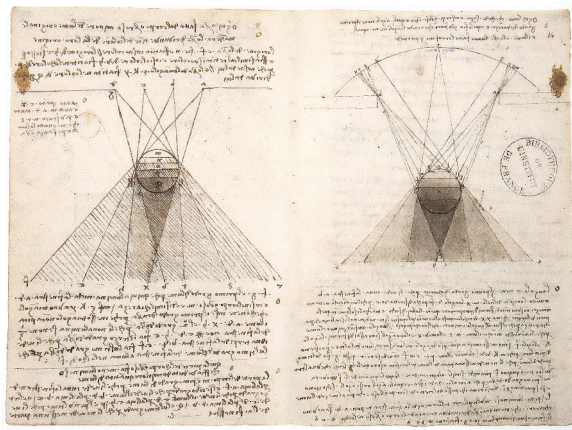


Figure 200: *Study of the Graduations of Shadows on Spheres* from Leonardo's *Codex Ashburnham*

CLASSICAL CHINESE AND JAPANESE ART made scarce use of shadow. *Six Persimmons* by the 13th century monk Mu Chi'I is a centerpiece of Zen art. The artist uses shade without shadows. No shadows for modeling. No cast shadows. Each persimmon is a simple silhouette, each singular in shade.



Mu Ch'I's Persimmons

By Gary Snyder

*There is no remedy for satisfying hunger
other than a painted rice cake.*

—Dōgen, November, 1242.

On a back wall down the hall

lit by a side glass door

is the scroll of Mu Ch'i's great
sumi painting, "Persimmons"

The wind-weights hanging from the
axles hold it still.

The best in the world, I say,
of persimmons.

Perfect statement of emptiness
no other than form
the twig and the stalk still on,
the way they sell them in the
market even now.

The original's in Kyoto at a
lovely Rinzai temple where they
show it once a year

this one's a perfect copy from Benrido
I chose the mounting elements myself
with the advice of the mounter

I hang it every fall.

And now, to these overripe
persimmons
from Mike and Barbara's orchard.
Napkin in hand,
I bend over the sink
suck the sweet orange goop
that's how I like it
gripping a little twig

those painted persimmons

sure cure hunger

Figure 201: *Six Persimmons* by Mu Chi'I, 13th century. [Link to Wikipedia](#)

THE MANNERIST style followed upon the early Italian Renaissance. Mannerism was a direct reaction to da Vinci, Rafael, Michaelangelo and others who strove for symmetry, proportion, and idealized beauty. Mannerists wanted to capture realism, but enhanced for dramatic effect.

GIOVANNI BATTISTA MORONI (~1520 - 1578) used his *Portrait of a Gentleman* to paint an essay on the realistic use of shadow. Here, he studies and compares shadows – a wide variety of cast shadows and modeling shadows – on convex and concave surfaces (Fig. 202).



Figure 202: *Portrait of a Gentleman*, Giovanni Battista Moroni, 1556. [National Gallery](#)

MICHELANGELO CARAVAGGIO (1571-1610) was the most influential Mannerist in using shadow. His followers were called the *tenebrists*, meaning shadowists. Where earlier Renaissance painters sought idealized and sanitized visual truth, Caravaggio insisted on absolute and brutal honesty. His painting of Thomas, the apostle who voiced skepticism about Jesus's resurrection, is a fully realistic depiction of one Bible verse:

Then saith he to Thomas, Reach hither thy finger, and behold my hands; and reach hither thy hand, and thrust it into my side: and be not faithless, but believing.

– John 20:27 (KJV)

CARAVAGGIO creates harsh shadows with dramatic and klieg-like intensity. The scene strives for earthly and tangible naturalism. The holy figures – Jesus, Thomas, and the other apostles – are painted as ordinary humans, worried men with wrinkled brows. Thomas pokes a real finger into a real wound. Caravaggio revolutionized religious depiction by populating biblical scenes with real people.



Figure 203: *The Unbelieving Thomas*, Caravaggio, 1601. [Sarassouci Museum](#)

ITALIAN REALISM made its way to Northern Europe. Rembrandt admired Caravaggio and the tenebrists. A painting from Rembrandt's school is more about shadow than light. Raking light streams through a window and creates an anamorphic shadow stretching across the wall.



Figure 204: *Man Seated Reading at a Table in a Lofty Room*. by Follower of Rembrandt or his school, 1628/30. [Link to painting at the National Gallery.](#)

PIETER JANSSENS ELINGA (1623-1682) was a contemporary of Vermeer. Elinga was as fascinated by shadow as Vermeer was fascinated by light. In this quiet domestic scene, a lady sits at a table, a maid sweeps the floor, and a painter stands in a back room. Shadows – from both direct and reflected light – achieve a complex and virtuosic display of realism.



Figure 205: *Interior with Painter, Woman Reading and Maid Sweeping*, by Pieter Janssens Elinga, 1665-1670. [Link to painting at the Städel Museum, Frankfurt](#)

THE SILHOUETTE was at the beginning of Western Art. But in the United States in the 18th and 19th centuries, it suddenly evolved into a ubiquitous and unique art form. Silhouettes can always be made with scissors and skilled hands. But American technological know-how created gadgets that manufactured high-quality silhouettes at high-speed. Any amateur could make a convincing silhouette. Any customer could afford the ennobling look of a Roman hero.

CHEMICAL PHOTOGRAPHY would end the brief but surging popularity of the American portrait silhouette. But the silhouette retains a timeless quality, and a perspective on other art forms by comparison. When an image is flattened and turned into black and white, all sense of depth and color are lost. The remaining silhouette keeps a sense of realism as pure shadow.

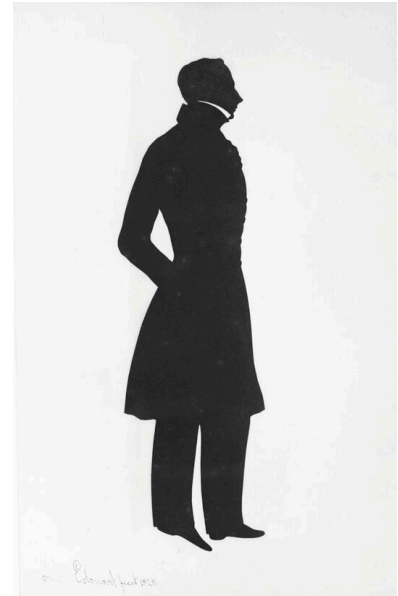


Figure 206: Augustin Amant Constant Fidèle Edouart, French, *Album of Silhouettes*. Harvard Art Museums.

Figure 207: [A Sure and Convenient Machine for Drawing Silhouettes](#) by Thomas Holloway ca. 1788.

THE PHYSIOGNOTRACE by John Hawkins was the major accelerator of silhouette mass-production. With the physiognotrace, anyone could trace a silhouette profile. The silhouette became a democratizing form of art, easy to afford, easy to make. Anyone from American citizen to American President could have a silhouette. The silhouette both equalized and ennobled, art that resonated with the mythic story of American democracy.

SILHOUETTE, a French word that is also connoted with elegance, actually had a less-than-courtly provenance. Étienne de Silhouette (1709-1761) was a French minister famous for being cheap. This elegant but economical form of art was named after a miser.

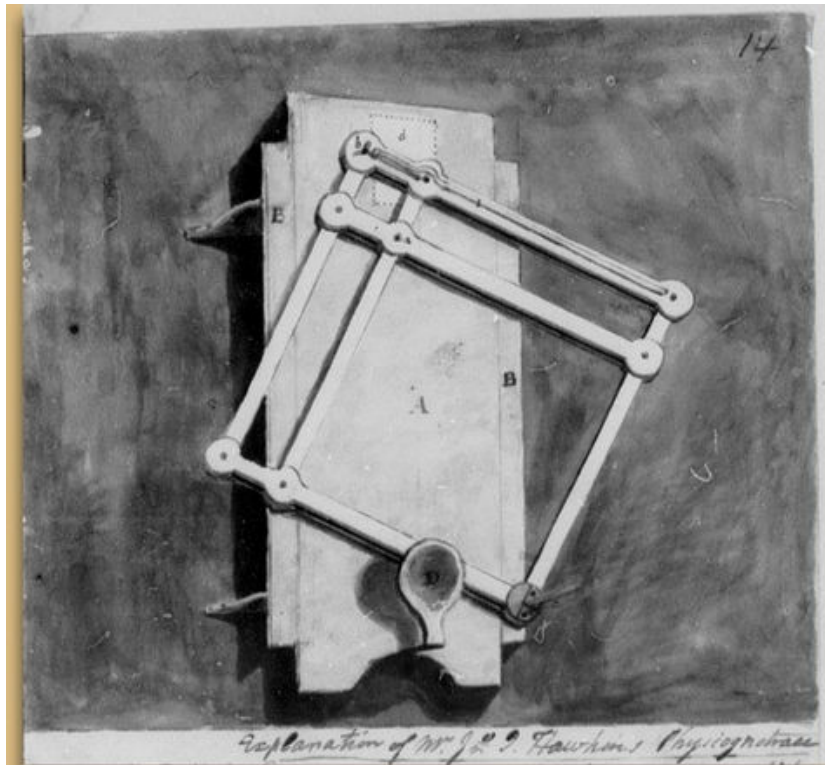


Figure 208: Hollow cut silhouette portrait of George Washington (ca. 1798), by Eleanor Custis, his adopted step-daughter and granddaughter of Martha Washington. [Link to Metropolitan Museum of Art.](#)



Figure 209: Jefferson Portrait by Raphaele Peale 1798. [Monticello Museum.](#)

Figure 210: Drawing of John Hawkin's physiognotrace device.

PHYSIOGNOMIC THEORIES – racism couched in the scientific language of ‘phrenology’ – held that the profile and shape of the skull was an indicator of intelligence and character. A less respectable driver of the popularity of the silhouette was the mistaken idea that it recorded more than a shadow, a rationale of debunked scientific theory.

LA AMISTAD was the slave ship that saw revolt in 1839. It was captured in New York and the case of the slaves was decided in *United States vs. The Amistad* in favor of freedom. However, in John Warner Barber’s *History of the Amistad Captives*, the identities of the 36 captives is shown only in grotesquely reductive silhouette, along with words evoking phrenology.

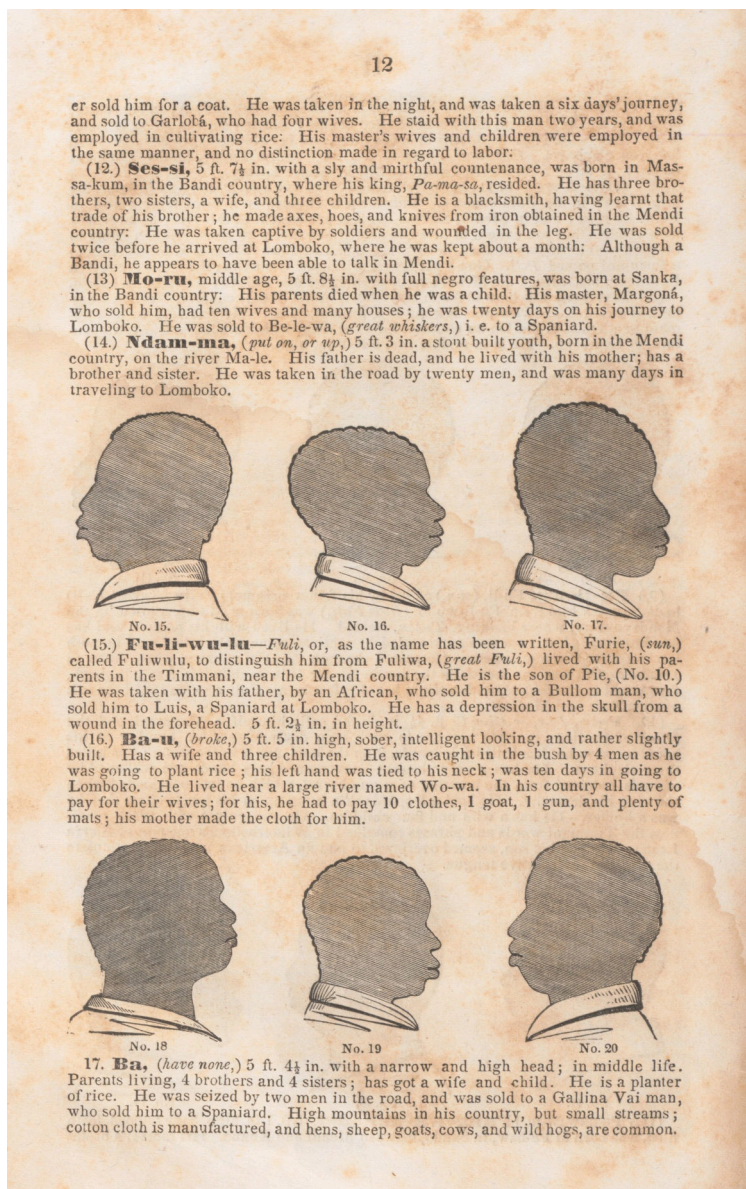


Figure 211: *A History of the Amistad Captives: Being a Circumstantial Account of the Capture of the Spanish Schooner Amistad, by the Africans on Board; Their Voyage, and Capture near Long Island, New York; with Biographical Sketches of Each of the Surviving Africans.* by John Warner Barber, 1840. [Metropolitan Museum](#)

A SILHOUETTE PROFILE can capture identifying detail with precision, but it is also unavoidably anonymizing and ambiguous. When a silhouette is made from any direction but side profile, identity is fully erased. *Skeuomorph* is the name of the iconic blank head that awaits photographs in social media “profiles”.

THE AESTHETIC OF ANONYMITY, using a silhouette not painted in profile, was used by artists. Gerard ter Borch (1617-1681), a contemporary of Vermeer, paints an unknown and unknowable soldier in his grim *Man on Horseback*.



Figure 212: Anonymous



Figure 213: *Man on Horseback* by Gerard ter Borch, 1634. [MFA](#)

CASPAR DAVID FRIEDRICH (1774-1840) painted his *Wanderer Above a Sea of Fog* as a silhouetted figure against a stormy landscape. Here, the anonymity of the silhouette has a more sublime purpose, letting any viewer take the part of the Wanderer. Of painting, Friedrich said:

The painter should paint not only what he has in front of him, but also what he sees inside himself. If he sees nothing within, then he should stop painting what is in front of him.



Figure 214: *Wanderer Above a Sea of Fog* by Caspar David Friedrich, 1817.
[Hamburger Kunsthalle](#)

MODERN ARTISTS have reinterpreted the silhouette as an art form. Kara Walker (1969-) manipulates the history of the silhouette in her disturbing and striking social commentaries (Fig. 216). In contrast to polite portraiture, what the silhouette largely was in the 19th century, Walker evokes exploitation and dark fantasy in 19th century life with her modern use of the art form.



Figure 215: *African/American* by Kara Walker, 1998. [Whitney Museum](#)



Figure 216: *Auntie Walkers Wall Sampler for Savages* by Kara Walker. Kara Walker, a contemporary artist, a mural-scale silhouettes that enacts the cruelty of antebellum plantation life.

KUMI YAMASHITA (1968-) plays with the dimensionality of the silhouette by shaping paper to create silhouettes as shadows when properly illuminated with raking light(Fig. 217). Yamashita's works are anamorphic shadows. The shadows make you think about how they were made.

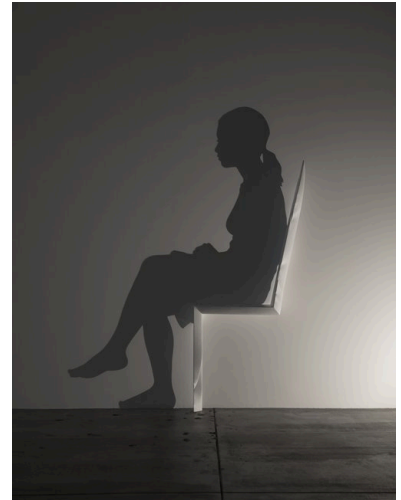


Figure 217: *Chair*, by Kumi Yamashita, 2014. Carved wood, single light source, shadow.

Figure 218: *Fragments*, by Kumi Yamashita, 2009.

FRANCESCA BEWER, the Director of the M-Lab at the Harvard Art Museums, made a surprisingly complex shadow in her photographic still life of lemons.



Figure 219: *Still Life with Oranges* by Francesca Bewer, 2024.

PHOTOGRAPHING

Taking Francesca's photograph as an inspiration, use your phone cameras to capture a surprising or intriguing shadow effect, whether something you stumble upon or something you arrange. Use words to describe what is happening in your photograph.

READING

- "Shades of Black and White" by Penley Knipe. From Asma Naeem. *Black Out: Silhouettes Then and Now*. Princeton University Press, Princeton, New Jersey, 2018. ISBN 069118058X. [Download paper](#)

BIBLIOGRAPHY

- E. H. (Ernst Hans) Gombrich. *Shadows : the depiction of cast shadows in western art*. Yale University Press, New Haven, new edition. edition, 2014. ISBN 9780300210040

Updated: April 7, 2025

WEEK TEN - DRAWING

Thursday, 10 April 2025, 12:45 - 2:15 PM EST.

Visit Ethan Murrow's Studio

59 Wareham Street Boston, MA 02118

DRAWING is the act of pulling an object towards oneself. When the object leaves a mark on a surface, it creates a 'drawing'. The earliest drawings were scratches on surfaces. An engraved stone slab from 11,000 BCE, found in France during the Palaeolithic, depicts part of a horse with a long tail. This plaquette – about 10 cm wide and 3 cm thick – was meant to be portable, shared, and seen.

THE NAZCA LINES in the Peruvian desert are among the largest ancient drawings, over 700 pictures of geometric shapes and animals. The largest are over 300 m, scratched into desert sand between 500 BC and 500 AD. These could not be seen by the artists who made them. Now, they can be seen by airplane or drone camera. They would have required painstaking planning and precision. Presumably they were drawn for the eyes of gods above.

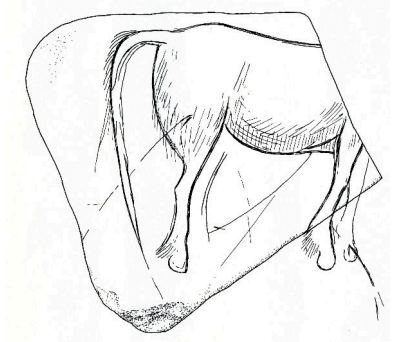


Figure 220: Engraved horse, [British Museum](#)



Figure 221: *The Spider* [Link to Wikipedia](#)

THE INVENTION OF PARCHMENT AND PAPER led to easily portable drawing surfaces and notebooks. At first parchment and paper was rare and expensive. The precious material was reserved for religious worship and text that was “illuminated” by drawings. Pictures were expressions of piety and added visual interest.

THE PRINTING PRESS made paper both widely available and affordable. Soon after Gutenberg’s printing press was reported in 1439, books began to be printed with movable type. Pictures began to be printed with engraved woodblocks and metal plates. Mass production made paper cheap, plentiful, and disposable.



Figure 222: Woodcut showing printer on left removing a page from the press while the printer on right inks a text block. Two printers could produce 3,600 pages a day.

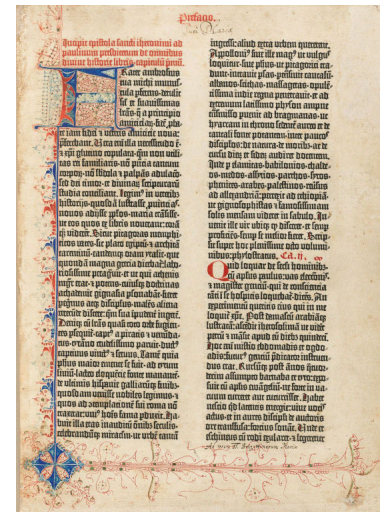


Figure 223: Page from Harvard’s copy of the Gutenberg Bible, printed on paper, one of 23 complete surviving copies. It is always on display in the Harry Elkins Widener Memorial Rooms.

Figure 224: The Utrecht Psalterbook was made in the ninth century, a masterpiece with 166 drawings in pen that accompany the psalms and texts. [Wikipedia](#)

WHEN PAPER BECAME PLENTIFUL, it became an efficient way of recording and working out ideas. Paper changed how artists worked, how ideas evolved, and how artistic innovation was communicated. With plentiful paper, Renaissance artists had ample, cheap material to practice and hone their skills. Simply being able to practice was a key turning point in Renaissance accuracy and realism. Before painting or sculpting a work of art, an artist could think, plan, and design with drawing.

PAOLO UCCELLO (1397-1475) worked out ideas about linear perspective on paper. Vasari wrote about Uccello his *Lives of the Most Excellent Painters, Sculptors, and Architects*, a solitary and single-minded man:

gifted with a sophisticated mind, he enjoyed investigating complex mechanisms and the strange products of the art of perspective.

His obsessiveness appears in his calculated drawing of a chalice. But Uccello, like many other artists, considered such drawings as preparatory and disposable. Most are lost.

THE NOTEBOOK was among Leonardo da Vinci's most powerful ideas. We have about 6000 sheets, about a fifth of the estimated total. On these paper sheets, he worked out ideas for paintings like the *Last Supper*. He recorded thoughts and discoveries like the natural flow of water. He tried every drawings tool that was available: pen and ink, metalpoint, charcoal, and chalk.



Updated: April 7, 2025

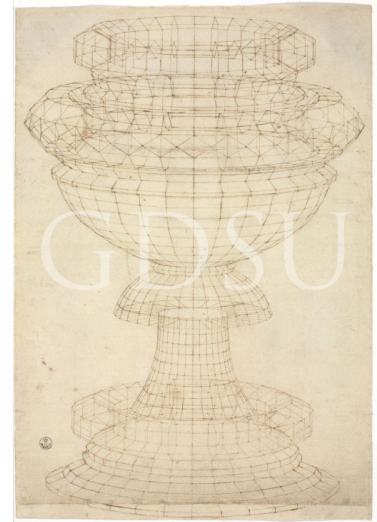


Figure 225: *Studio prospettico di calice*, 1450-70, Paolo Uccello. [Uffizzi](#)



Figure 226: Studies for *The Adoration of the Magi* and *The Last Supper* in brown ink with quill pen and leadpoint, Leonardo da Vinci, [Louvre](#)

Figure 227: *A deluge*, Black chalk, pen and ink, wash. 1517-1518, Leonardo da Vinci. [Royal Collection Trust](#)

MICHAELANGELO BUONAROTTI (1475-1564) started his paintings with drawings, which he usually burned. Vasari writes

to the end that no one might see the labours endured by him and his methods of trying his genius, and that he might not appear less than perfect.

About 600 sheets survive, including studies for the Sistine Chapel. For Adam, he carefully planned his musculature until he found the pose that he wanted. This drawing incorporated Michaelangelo's deep knowledge of the human body, learned from studies of flayed corpses. Michaelangelo's studies led to Adam's convincing pose. This pose was invented from his studies, not drawn from life. It is physically impossible for a human to bend their torso so sharply. These drawings were then enlarged on tracing paper and transferred to the ceiling before Michaelangelo began painting.

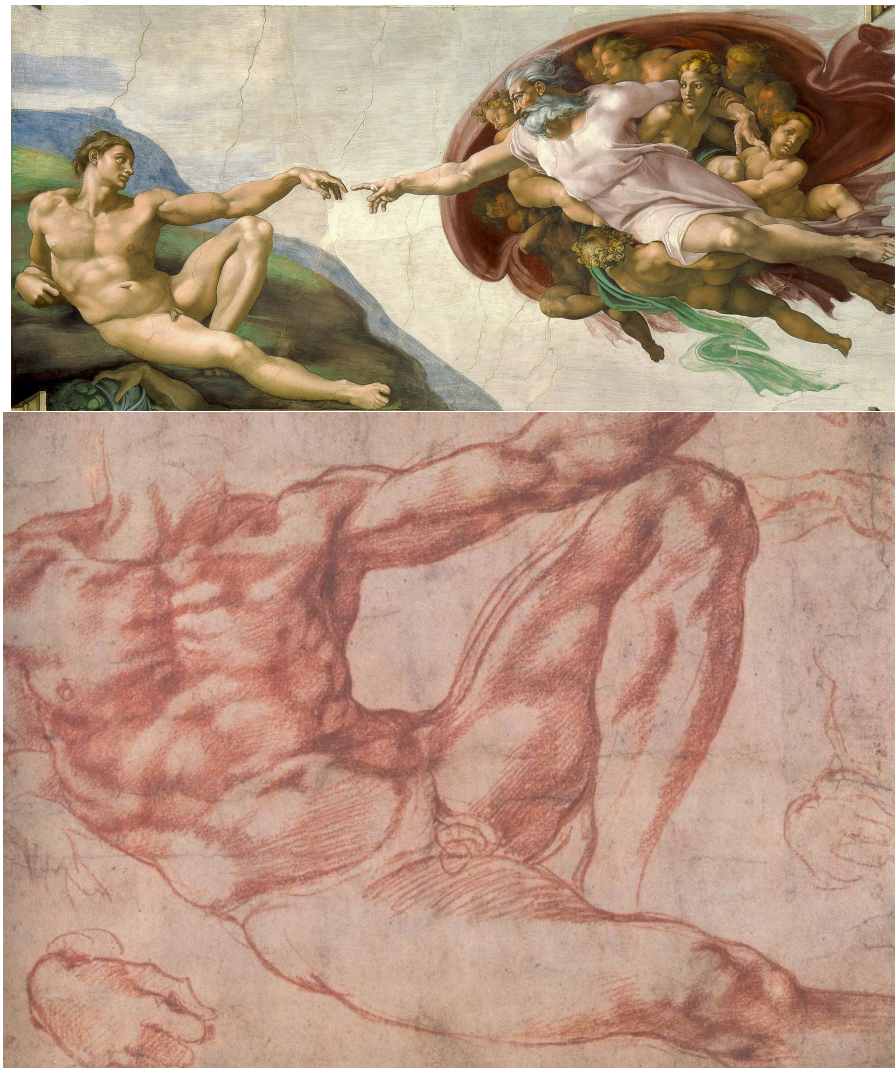


Figure 228: *Creation of Adam* by Michelangelo, 1512 and red chalk study.

SIR PETER PAUL RUBENS (1577-1640) painted grand panoramic scenes that also did not occur. His landscapes were built from ideas recorded on paper as he walked the Flemish countryside. His panel painting of *Het Steen*, the manor where he lived, invents an autumn morning. The painting draws on notes on paper, from drawings and sketches were he recorded effects like the play of light on water. In one drawing, he records an optical phenomenon that he wanted to remember:

The reflection of the trees in the water is browner and more perfect in the water than the trees themselves.



Figure 229: *An Autumn Landscape with a View of Het Steen in the Early Morning* by Rubens, 1636 [National Gallery](#).



Figure 230: *Trees reflected in water at sunset* by Rubens, 1635-38. Pen and brown ink. [British Museum](#)

DRAWING TOOLS IMPROVED OVER TIME. Renaissance tools like ink and metalpoint are unforgiving, impossible to erase. Chalk and charcoal smudged easily. When the graphite pencil was invented in the late 18th century, artists had a new and powerful tool. The point was easily sharpened. Faint and hard lines, texture and shading, could be made with varying pressure and different strokes. The pencil helped turn pure drawings into finished works of art, not just as a step toward other final form like a print or painting.

JOHN CONSTABLE is known for oil landscapes. But his exquisitely detailed graphite drawing of elm trees is a fully finished work of art. To other drawings, he sometimes added watercolor. His attitude to these landscapes, across mediums, is summed up in his quote:

Painting is a science, and should be pursued as an inquiry into the laws of nature. Why, then, may not landscape painting be considered as a branch of natural philosophy, of which pictures are but the experiments?



Figure 231: *A Mill Near Colchester*, 1833, John Constable Watercolor over traces of graphite, with scratching out and a touch of pen and brown ink on the horizon, evidence of squaring for transfer in graphite, with a color wash test verso [Metropolitan Museum](#)

Figure 232: *Elm trees in Old Hall Park, East Bergholt*, 1817, John Constable, graphite on paper. [Victoria and Albert Museum](#)



Figure 233: *A Landscape with a Church Tower and a House* 1809, John Constable, Graphite on cream antique laid paper. [Harvard Art Museums](#)

MANY MODERN ARTISTS now use drawing as their main medium. Ethan Murrow makes large-scale drawings on paper with pencil and murals with paint-pencils. In his work, he looks back on historical narratives, and the ways in which stories are told. Murrow is an admirer of the exquisite naturalistic detail of Jan Van Eyck. In *Passage*, he draws himself boating by a lamb, a quote from *The Ghent Altarpiece*, also called *The Adoration of the Mystic Lamb*.



Figure 234: Images from the Expedia Mural Project in 2019 and from "Hauling" commissioned by the Currier Museum in 2018, by Ethan Murrow.

Updated: April 7, 2025

Figure 235: *Passage*, 2023, Ethan Murrow, graphite on paper, 48"x48".

GIDEON BOK draws the passage of time in confined spaces. He depicts his own his own studio, over and over in various states at different times. By adding a temporal dimension, he makes the room seem alive (Archives of askArt.)



Updated: April 7, 2025

TOBA KHEDOORI often depicting objects from different perspectives. By isolating objects in an unfamiliar setting, she hovers between naturalistic representation and abstraction. Her work is painstaking and precise, slowly working by hand (from www.lacma.org)



Updated: April 7, 2025

PETER WAITE puts structures that we have built into new perspectives. He makes large scale drawings of recognizable places: stadiums, gardens, bridges, museums, monuments, train stations, schools, prisons, casinos, and temples. These are spaces to which we form specific attachments. Looking at these spaces in drawings shifts our perspective. (from www.peterwaite.com)



Updated: April 7, 2025

WATCHING AND WRITING

Watch the movie *Tim's Vermeer*, freely available on the course website or hosted by Claire Swadling. It was by seeing Tim's Vermeer that I first wondered how realistic artworks were made, the first inspiration for inventing this course. We used to watch it at the beginning of the class. Instead, we're watching it at the end. In this week's essay, react to the movie with what you now know.

BIBLIOGRAPHY

- Susan Owens. *The story of drawing : an alternative history of art*. Yale University Press, New Haven, 2024. ISBN 9780300277609



Figure 236: *Two Women*, c. 1878/1880, by Edgar Degas, pastel over watercolor. Degas was an oil painter, but as he got older, he drifted toward drawing with pastels. Painting with oil grew tedious, and pastels gave him the strength of painting with the ease of drawing.
[National Gallery](#)